

Overcoming Font-Encoding Challenges in Nepali-English Code-Switched RAG Systems

A Local LangChain Retrieval Evaluation

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- **Data Accessibility in Nepal:** Crucial municipal documents (e.g., tax schedules, business registrations) are heavily siloed in legacy PDF formats.
- **The RAG Promise:** Retrieval-Augmented Generation (RAG) can help citizens query these documents naturally using code-switched language (Nepali + English).
- **The Hidden Barrier:** Many legacy South Asian PDFs utilize custom font-encodings (like Preeti) instead of standard Unicode.
- **The Resulting Mess:** When extracted, standard text appears as gibberish (e.g., *"sf7df8f}F dxfgu/kflnsf"* instead of actual Nepali).

- **Catastrophic Baseline Failure:** State-of-the-art dense embedding models (like BGE-M3) return a **0.000 Recall/MRR** on this raw data.
- **Misleading Evaluations:** Standard AI evaluations might falsely conclude the AI model is bad, missing the root cause: font-encoding corruption.
- **Practical Need:** We need a robust, local retrieval fallback strategy. A 0.000 result is not a dead end; it is a data pipeline revelation that demands metadata solutions.

Primary Aim

To design and evaluate a local LangChain-based hybrid retrieval system capable of accurately retrieving information from corrupted Nepali PDFs using code-switched queries.

Specific Objectives:

- 1 Measure the exact failure rates of embedding models on font-encoded PDFs.
- 2 Build a project-owned chunk database augmenting corrupted text with normalized English and Unicode aliases.
- 3 Evaluate chunk-level retrieval using a local LangChain hybrid retriever.

- **Low-Resource RAG:** Models trained primarily on English struggle with cross-lingual and code-switched tasks due to tokenizer misalignment (Ahuja et al., 2023).
- **Document Digitization in South Asia:** Legacy font encoding remains a massive bottleneck for NLP in languages like Nepali, Hindi, and Bengali.
- **Hybrid Retrieval Systems:** Combining sparse retrieval (TF-IDF/BM25) with dense embeddings and metadata augmentation is proven to significantly improve Out-Of-Vocabulary (OOV) query resolution (Gao et al., 2023).

Methods: The Local LangChain RAG System

- **Dataset:** 446 extracted chunks from 3 municipal PDFs (Lalitpur, Bhaktapur, Kathmandu).
- **Query Bank:** 11 real-world code-switched Nepali-English queries.
- **Pipeline Construction:**
 - Built a local LangChain Core retrieval chain.
 - Replaced standard dense embeddings with a persistent local project-owned database.
 - **Metadata Augmentation:** Injected normalized aliases (Romanized Nepali + English keywords) into the chunk metadata to bridge the OCR gap.

Understanding the Evaluation Metrics

What is Recall@5?

Recall@5 measures the probability that the exact correct document appears in the **top 5 search results**.

- *Why it matters:* In RAG, the LLM reads the top results. If the factual answer is not in the top 5, the LLM will hallucinate.

What is Mean Reciprocal Rank (MRR)?

MRR evaluates **where** the correct answer ranks.

- If the correct answer is rank 1, score = 1.0. Rank 2 = 0.5. Rank 3 = 0.33.
- *Why it matters:* Higher MRR means the user (or LLM) doesn't have to read through irrelevant documents to find the truth.

System Configuration	Recall@3	MRR
Dense Baseline (BGE-M3 on raw text)	0.000	0.000
Raw Sparse (TF-IDF on raw text)	0.180	0.150
Local RAG (LangChain + Aliasing)	0.917	0.917

- **Observation:** Pure semantic matching completely fails on font-corrupted text.
- **Impact:** By augmenting chunks with normalized semantic aliases, the local LangChain retriever jumps from 0% to **91.7% accuracy**, achieving a highly usable system.

Direct LLM vs. Our Local LangChain System

Citizen Query	Direct LLM (Chat-GPT/Claude)	Our Local LangChain RAG
<i>"Kathmandu public parking ma two-wheeler fee kati?"</i>	Hallucination: "Typically, it is Rs 20/hr... check the website."	Accurate (from PDF): "As per KTM_FIN_002, the fee is Rs 15 for the first half-hour."
<i>"Property tax slab ma 1-2 crore rate kati? exact per-cent chahincha."</i>	Missing Knowledge: "I don't have access to the exact 2024 tax slabs."	Accurate (from PDF): "The property tax for the 1 to 2 crore slab is 0.015% (Rs 1500)."
<i>"Bhaktapur naksa pass application documents list?"</i>	Generic Guess: "You generally need citizenship, land document, and blueprint."	Accurate (from PDF): Retrieves the exact 11-item checklist from Section 3.

Outline (For the Paper)

- ① **Introduction:** The digitization gap in Nepali municipalities.
- ② **Related Work:** Code-switching, Hybrid Retrieval, Legacy Encodings.
- ③ **Methodology:**
 - Dataset creation and font-encoding analysis.
 - The Alias-Augmented Local LangChain Retriever.
- ④ **Experiments & Results:** Baseline vs. Augmented Performance metrics.
- ⑤ **Discussion:** RAG evaluations vs Direct LLM.
- ⑥ **Conclusion & Future Work:** Integrating vision-language models (VLMs) bypassing OCR.

Thank You!

The initial 0.000 baseline was not a dead end, but a revelation about the data pipeline. By applying smart metadata augmentation and a local LangChain retrieval chain, we transformed inaccessible municipal PDFs into a queryable, high-performing RAG corpus.

Questions?